In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sqlite3
```

<h3 id="1.-Import-Database-/-Data">1. Import Database / Data</h3>

In [2]:
```
conn = sqlite3.connect('customer_churn.db')

sql_query = """ select name from sqlite_master where type = 'table'; """

tables = pd.read_sql(sql_query, conn)

# Create dataframe for each table

for table_name in tables['name']:
    df = pd.read_sql(f"select * from {table_name}", conn)
    globals()[f"df_{table_name}"] = df
    print(f"Created Dataframe: df_{table_name}")

conn.close()
```

```
Created Dataframe: df_db_customer
Created Dataframe: df_db_subscription
Created Dataframe: df_db_support
```

In [3]:
```
# To check the column name in Database (Optional)

# Print Table names and Column names

conn = sqlite3.connect('customer_churn.db')

for table_name in tables['name']:
    print(f"\nTable Name: {table_name}")

    # Get column information
    columns_query = f"PRAGMA table_info({table_name});"
    columns = pd.read_sql(columns_query, conn)
    print("columns:")
    print(columns['name'].tolist())

conn.close()
```

```
Table Name: db_customer
columns:
['customerid', 'name', 'country', 'state', 'gender', 'dob', 'interests', 'pincod
e']

Table Name: db_subscription
columns:
['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'p
lan_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly_
charges', 'cltv', 'churn_score']

Table Name: db_support
columns:
['customerid', 'complaint_date', 'escalations', 'csat_score', 'col_1', 'comment']
```

<h3 id="2.-Data-Cleaning">2. Data Cleaning</h3>

In [4]:
```
df_db_customer.head()
```

Out[4]:

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> <th>interests</th> <th>pincode</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0002-ORFBO | keshav | India | Maharashtra | Male | 1982-04-12 00:00:00 | travel | None |
| 0003-MKNFE | raghav | India | Karnataka | Male | 1995-11-23 00:00:00 | NaN | None |
| 0004-TLHLJ | lalita | India | Delhi | Female | 1978-02-15 00:00:00 | movie | None |
| 0011-IGKFF | mohan | India | Nagaland | Male | 2001-08-30 00:00:00 | NaN | None |
| 0013-EXCHZ | mira | India | Delhi | Female | 1990-05-05 00:00:00 | drama | None |

In [5]:
```
df_db_customer.tail()
```

Out[5]:

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> <th>interests</th> <th>pincode</th> </thead> <tbody> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0020-JDNXP | rikim | India | Meghalaya | Female | 1994-08-19 00:00:00 | NaN | None |
| 0021-IKXGC | vishakha | India | Rajasthan | Female | 2000-09-02 00:00:00 | NaN | None |
| 0022-TCJCI | raghvendra | India | Telangana | Male | 1983-12-30 00:00:00 | NaN | None |
| 0023-HGHWL | rishabh | India | Uttar Pradesh | Men | 1991-05-14 00:00:00 | NaN | None |
| 0023-UYUPN | sudevi | India | Maharashtra | Women | 1977-10-06 00:00:00 | NaN | None |

In [6]:
```
df_db_customer.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   customerid  21 non-null     str
 1   name        21 non-null     str
 2   country     18 non-null     str
 3   state       21 non-null     str
 4   gender      21 non-null     str
 5   dob         21 non-null     str
 6   interests   4 non-null      str
 7   pincode     0 non-null      object
dtypes: object(1), str(7)
memory usage: 1.4+ KB
```

In [7]:
```
# A. Rename Column - name

df_db_customer.rename(columns = {'name' : 'customer_name'}, inplace = True)
```

In [8]:
```
df_db_customer
```

Out[8]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> <th>interests</th> <th>pincode</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0002-ORFBO | keshav | India | Maharashtra | Male | 1982-04-12 00:00:00 | travel | None |
| 0003-MKNFE | raghav | India | Karnataka | Male | 1995-11-23 00:00:00 | NaN | None |
| 0004-TLHLJ | lalita | India | Delhi | Female | 1978-02-15 00:00:00 | movie | None |
| 0011-IGKFF | mohan | India | Nagaland | Male | 2001-08-30 00:00:00 | NaN | None |
| 0013-EXCHZ | mira | India | Delhi | Female | 1990-05-05 00:00:00 | drama | None |
| 0013-MHZWF | durga | NaN | Delhi | Women | 1988-12-10 00:00:00 | NaN | None |
| 0013-SMEOE | mina | India | Meghalaya | Female | 1976-09-21 00:00:00 | NaN | None |
| 0014-BMAQU | madan | India | Rajasthan | Male | 1999-03-14 00:00:00 | NaN | None |
| 0015-UOCOJ | maya | NaN | Kathmandu | Women | 1985-07-07 00:00:00 | NaN | None |
| 0016-QLJIS | arjun | Nepal | Kathmandu | Male | 1993-10-29 00:00:00 | NaN | None |
| 0017-DINOC | shiva | India | Maharashtra | Men | 1997-01-22 00:00:00 | NaN | None |
| 0017-IUDMW | rangadevi | India | Karnataka | Female | 1981-06-18 00:00:00 | NaN | None |
| 0018-NYROU | chitra | NaN | Telangana | Female | 2004-12-01 00:00:00 | NaN | None |
| 0019-EFAEP | raju | India | Meghalaya | Female | 1992-04-25 00:00:00 | NaN | None |
| 0019-GFNTW | Madhav | India | Uttar Pradesh | Men | 1979-11-11 00:00:00 | NaN | None |
| 0020-INWCK | parvati | India | Delhi | Female | 1986-02-28 00:00:00 | job | None |
| 0020-JDNXP | rikim | India | Meghalaya | Female | 1994-08-19 00:00:00 | NaN | None |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0021-IKXGC | vishakha | India | Rajasthan | Female | 2000-09-02 00:00:00 | NaN | None |
| 0022-TCJCI | raghvendra | India | Telangana | Male | 1983-12-30 00:00:00 | NaN | None |
| 0023-HGHWL | rishabh | India | Uttar Pradesh | Men | 1991-05-14 00:00:00 | NaN | None |
| 0023-UYUPN | sudevi | India | Maharashtra | Women | 1977-10-06 00:00:00 | NaN | None |

In [9]:
```
# B. Drop columns - Interests and pincode

# Option 1

df_db_customer.drop(df_db_customer.columns[6:], axis = 1)              # Axis = 1
```

Out[9]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | | | |
|---|---|---|---|---|---|
| 0002-ORFBO | keshav | India | Maharashtra | Male | 1982-04-12 00:00:00 |
| 0003-MKNFE | raghav | India | Karnataka | Male | 1995-11-23 00:00:00 |
| 0004-TLHLJ | lalita | India | Delhi | Female | 1978-02-15 00:00:00 |
| 0011-IGKFF | mohan | India | Nagaland | Male | 2001-08-30 00:00:00 |
| 0013-EXCHZ | mira | India | Delhi | Female | 1990-05-05 00:00:00 |
| 0013-MHZWF | durga | NaN | Delhi | Women | 1988-12-10 00:00:00 |
| 0013-SMEOE | mina | India | Meghalaya | Female | 1976-09-21 00:00:00 |
| 0014-BMAQU | madan | India | Rajasthan | Male | 1999-03-14 00:00:00 |
| 0015-UOCOJ | maya | NaN | Kathmandu | Women | 1985-07-07 00:00:00 |
| 0016-QLJIS | arjun | Nepal | Kathmandu | Male | 1993-10-29 00:00:00 |
| 0017-DINOC | shiva | India | Maharashtra | Men | 1997-01-22 00:00:00 |
| 0017-IUDMW | rangadevi | India | Karnataka | Female | 1981-06-18 00:00:00 |
| 0018-NYROU | chitra | NaN | Telangana | Female | 2004-12-01 00:00:00 |
| 0019-EFAEP | raju | India | Meghalaya | Female | 1992-04-25 00:00:00 |
| 0019-GFNTW | Madhav | India | Uttar Pradesh | Men | 1979-11-11 00:00:00 |
| 0020-INWCK | parvati | India | Delhi | Female | 1986-02-28 00:00:00 |
| 0020-JDNXP | rikim | India | Meghalaya | Female | 1994-08-19 00:00:00 |
| 0021-IKXGC | vishakha | India | Rajasthan | Female | 2000-09-02 00:00:00 |
| 0022-TCJCI | raghvendra | India | Telangana | Male | 1983-12-30 00:00:00 |
| 0023-HGHWL | rishabh | India | Uttar Pradesh | Men | 1991-05-14 00:00:00 |
| 0023-UYUPN | sudevi | India | Maharashtra | Women | 1977-10-06 00:00:00 |

In [10]:
```
# Option 2

df_db_customer.drop(df_db_customer.columns[-2:], axis = 1)
```

Out[10]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | | | |
|---|---|---|---|---|---|
| 0002-ORFBO | keshav | India | Maharashtra | Male | 1982-04-12 00:00:00 |
| 0003-MKNFE | raghav | India | Karnataka | Male | 1995-11-23 00:00:00 |
| 0004-TLHLJ | lalita | India | Delhi | Female | 1978-02-15 00:00:00 |
| 0011-IGKFF | mohan | India | Nagaland | Male | 2001-08-30 00:00:00 |
| 0013-EXCHZ | mira | India | Delhi | Female | 1990-05-05 00:00:00 |
| 0013-MHZWF | durga | NaN | Delhi | Women | 1988-12-10 00:00:00 |
| 0013-SMEOE | mina | India | Meghalaya | Female | 1976-09-21 00:00:00 |
| 0014-BMAQU | madan | India | Rajasthan | Male | 1999-03-14 00:00:00 |
| 0015-UOCOJ | maya | NaN | Kathmandu | Women | 1985-07-07 00:00:00 |
| 0016-QLJIS | arjun | Nepal | Kathmandu | Male | 1993-10-29 00:00:00 |
| 0017-DINOC | shiva | India | Maharashtra | Men | 1997-01-22 00:00:00 |
| 0017-IUDMW | rangadevi | India | Karnataka | Female | 1981-06-18 00:00:00 |
| 0018-NYROU | chitra | NaN | Telangana | Female | 2004-12-01 00:00:00 |
| 0019-EFAEP | raju | India | Meghalaya | Female | 1992-04-25 00:00:00 |
| 0019-GFNTW | Madhav | India | Uttar Pradesh | Men | 1979-11-11 00:00:00 |
| 0020-INWCK | parvati | India | Delhi | Female | 1986-02-28 00:00:00 |
| 0020-JDNXP | rikim | India | Meghalaya | Female | 1994-08-19 00:00:00 |
| 0021-IKXGC | vishakha | India | Rajasthan | Female | 2000-09-02 00:00:00 |
| 0022-TCJCI | raghvendra | India | Telangana | Male | 1983-12-30 00:00:00 |
| 0023-HGHWL | rishabh | India | Uttar Pradesh | Men | 1991-05-14 00:00:00 |
| 0023-UYUPN | sudevi | India | Maharashtra | Women | 1977-10-06 00:00:00 |

In [11]:
```
# Option 3

df_db_customer.drop(columns = ['interests', 'pincode'], inplace = True)
```

In [12]:
```
df_db_customer
```

Out[12]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | | | |
|---|---|---|---|---|---|
| 0002-ORFBO | keshav | India | Maharashtra | Male | 1982-04-12 00:00:00 |
| 0003-MKNFE | raghav | India | Karnataka | Male | 1995-11-23 00:00:00 |
| 0004-TLHLJ | lalita | India | Delhi | Female | 1978-02-15 00:00:00 |
| 0011-IGKFF | mohan | India | Nagaland | Male | 2001-08-30 00:00:00 |
| 0013-EXCHZ | mira | India | Delhi | Female | 1990-05-05 00:00:00 |
| 0013-MHZWF | durga | NaN | Delhi | Women | 1988-12-10 00:00:00 |
| 0013-SMEOE | mina | India | Meghalaya | Female | 1976-09-21 00:00:00 |
| 0014-BMAQU | madan | India | Rajasthan | Male | 1999-03-14 00:00:00 |
| 0015-UOCOJ | maya | NaN | Kathmandu | Women | 1985-07-07 00:00:00 |
| 0016-QLJIS | arjun | Nepal | Kathmandu | Male | 1993-10-29 00:00:00 |
| 0017-DINOC | shiva | India | Maharashtra | Men | 1997-01-22 00:00:00 |
| 0017-IUDMW | rangadevi | India | Karnataka | Female | 1981-06-18 00:00:00 |
| 0018-NYROU | chitra | NaN | Telangana | Female | 2004-12-01 00:00:00 |
| 0019-EFAEP | raju | India | Meghalaya | Female | 1992-04-25 00:00:00 |
| 0019-GFNTW | Madhav | India | Uttar Pradesh | Men | 1979-11-11 00:00:00 |
| 0020-INWCK | parvati | India | Delhi | Female | 1986-02-28 00:00:00 |
| 0020-JDNXP | rikim | India | Meghalaya | Female | 1994-08-19 00:00:00 |
| 0021-IKXGC | vishakha | India | Rajasthan | Female | 2000-09-02 00:00:00 |
| 0022-TCJCI | raghvendra | India | Telangana | Male | 1983-12-30 00:00:00 |
| 0023-HGHWL | rishabh | India | Uttar Pradesh | Men | 1991-05-14 00:00:00 |
| 0023-UYUPN | sudevi | India | Maharashtra | Women | 1977-10-06 00:00:00 |

In [13]:
```
df_db_customer.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   customerid     21 non-null     str
 1   customer_name  21 non-null     str
 2   country        18 non-null     str
 3   state          21 non-null     str
 4   gender         21 non-null     str
 5   dob            21 non-null     str
dtypes: str(6)
memory usage: 1.1 KB
```

In [14]:
```
# C. Change Data Type - dob

pd.to_datetime(df_db_customer['dob'])
```

Out[14]:
```
0    1982-04-12
1    1995-11-23
2    1978-02-15
3    2001-08-30
4    1990-05-05
5    1988-12-10
6    1976-09-21
7    1999-03-14
8    1985-07-07
9    1993-10-29
10   1997-01-22
11   1981-06-18
12   2004-12-01
13   1992-04-25
14   1979-11-11
15   1986-02-28
16   1994-08-19
17   2000-09-02
18   1983-12-30
19   1991-05-14
20   1977-10-06
Name: dob, dtype: datetime64[us]
```

In [15]:
```
df_db_customer['dob'] = pd.to_datetime(df_db_customer['dob'])
```

In [16]:
```
df_db_customer
```

Out[16]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | | | |
|---|---|---|---|---|---|
| 0002-ORFBO | keshav | India | Maharashtra | Male | 1982-04-12 |
| 0003-MKNFE | raghav | India | Karnataka | Male | 1995-11-23 |
| 0004-TLHLJ | lalita | India | Delhi | Female | 1978-02-15 |
| 0011-IGKFF | mohan | India | Nagaland | Male | 2001-08-30 |
| 0013-EXCHZ | mira | India | Delhi | Female | 1990-05-05 |
| 0013-MHZWF | durga | NaN | Delhi | Women | 1988-12-10 |
| 0013-SMEOE | mina | India | Meghalaya | Female | 1976-09-21 |
| 0014-BMAQU | madan | India | Rajasthan | Male | 1999-03-14 |
| 0015-UOCOJ | maya | NaN | Kathmandu | Women | 1985-07-07 |
| 0016-QLJIS | arjun | Nepal | Kathmandu | Male | 1993-10-29 |
| 0017-DINOC | shiva | India | Maharashtra | Men | 1997-01-22 |
| 0017-IUDMW | rangadevi | India | Karnataka | Female | 1981-06-18 |
| 0018-NYROU | chitra | NaN | Telangana | Female | 2004-12-01 |
| 0019-EFAEP | raju | India | Meghalaya | Female | 1992-04-25 |
| 0019-GFNTW | Madhav | India | Uttar Pradesh | Men | 1979-11-11 |
| 0020-INWCK | parvati | India | Delhi | Female | 1986-02-28 |
| 0020-JDNXP | rikim | India | Meghalaya | Female | 1994-08-19 |
| 0021-IKXGC | vishakha | India | Rajasthan | Female | 2000-09-02 |
| 0022-TCJCI | raghvendra | India | Telangana | Male | 1983-12-30 |
| 0023-HGHWL | rishabh | India | Uttar Pradesh | Men | 1991-05-14 |
| 0023-UYUPN | sudevi | India | Maharashtra | Women | 1977-10-06 |

In [17]:
```
df_db_customer.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   customerid     21 non-null     str
 1   customer_name  21 non-null     str
 2   country        18 non-null     str
 3   state          21 non-null     str
 4   gender         21 non-null     str
 5   dob            21 non-null     datetime64[us]
dtypes: datetime64[us](1), str(5)
memory usage: 1.1 KB
```

In [18]:
```
# D. Data Standardization - Gender

df_db_customer['gender'].unique()
```

Out[18]:
```
<StringArray>
['Male', 'Female', 'Women', 'Men']
Length: 4, dtype: str
```

In [19]:
```
df_db_customer['gender'].replace({'Men' : 'Male', 'Women' : 'Female'})
```

Out[19]:
```
0       Male
1       Male
2     Female
3       Male
4     Female
5     Female
6     Female
7       Male
8     Female
9       Male
10      Male
11    Female
12    Female
13    Female
14      Male
15    Female
16    Female
17    Female
18      Male
19      Male
20    Female
Name: gender, dtype: str
```

In [20]:
```
df_db_customer['gender'] = df_db_customer['gender'].replace({'Men' : 'Male', 'Wo
```

In [21]:
```
df_db_customer
```

Out[21]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | | | |
|---|---|---|---|---|---|
| 0002-ORFBO | keshav | India | Maharashtra | Male | 1982-04-12 |
| 0003-MKNFE | raghav | India | Karnataka | Male | 1995-11-23 |
| 0004-TLHLJ | lalita | India | Delhi | Female | 1978-02-15 |
| 0011-IGKFF | mohan | India | Nagaland | Male | 2001-08-30 |
| 0013-EXCHZ | mira | India | Delhi | Female | 1990-05-05 |
| 0013-MHZWF | durga | NaN | Delhi | Female | 1988-12-10 |
| 0013-SMEOE | mina | India | Meghalaya | Female | 1976-09-21 |
| 0014-BMAQU | madan | India | Rajasthan | Male | 1999-03-14 |
| 0015-UOCOJ | maya | NaN | Kathmandu | Female | 1985-07-07 |
| 0016-QLJIS | arjun | Nepal | Kathmandu | Male | 1993-10-29 |
| 0017-DINOC | shiva | India | Maharashtra | Male | 1997-01-22 |
| 0017-IUDMW | rangadevi | India | Karnataka | Female | 1981-06-18 |
| 0018-NYROU | chitra | NaN | Telangana | Female | 2004-12-01 |
| 0019-EFAEP | raju | India | Meghalaya | Female | 1992-04-25 |
| 0019-GFNTW | Madhav | India | Uttar Pradesh | Male | 1979-11-11 |
| 0020-INWCK | parvati | India | Delhi | Female | 1986-02-28 |
| 0020-JDNXP | rikim | India | Meghalaya | Female | 1994-08-19 |
| 0021-IKXGC | vishakha | India | Rajasthan | Female | 2000-09-02 |
| 0022-TCJCI | raghvendra | India | Telangana | Male | 1983-12-30 |
| 0023-HGHWL | rishabh | India | Uttar Pradesh | Male | 1991-05-14 |
| 0023-UYUPN | sudevi | India | Maharashtra | Female | 1977-10-06 |

In [22]:
```
df_db_customer['gender'].unique()
```

Out[22]:
```
<StringArray>
['Male', 'Female']
Length: 2, dtype: str
```

In [23]:
```
# E. Fix missing values = Country

df_db_customer['country'].isna()
```

Out[23]:
```
0     False
1     False
2     False
3     False
4     False
5      True
6     False
7     False
8      True
9     False
10    False
11    False
12     True
13    False
14    False
15    False
16    False
17    False
18    False
19    False
20    False
Name: country, dtype: bool
```

In [24]:
```
df_db_customer[df_db_customer['country'].isna()]
```

Out[24]:

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>5</th> <th>8</th> <th>12</th> </tbody> <tbody></tbody>

| | | | | | | |
|---|---|---|---|---|---|---|
| 0013-MHZWF | durga | NaN | | Delhi | Female | 1988-12-10 |
| 0015-UOCOJ | maya | NaN | | Kathmandu | Female | 1985-07-07 |
| 0018-NYROU | chitra | NaN | | Telangana | Female | 2004-12-01 |

In [25]:
```
# Country and state - Unique value pair

state_country_mapping = df_db_customer.dropna(subset=['country']).set_index('sta
state_country_mapping
```

Out[25]:
```
{'Maharashtra': 'India',
 'Karnataka': 'India',
 'Delhi': 'India',
 'Nagaland': 'India',
 'Meghalaya': 'India',
 'Rajasthan': 'India',
 'Kathmandu': 'Nepal',
 'Uttar Pradesh': 'India',
 'Telangana': 'India'}
```

In [26]:
```
df_db_customer['country'] = df_db_customer['country'].fillna(df_db_customer['sta
```

In [27]:
```
df_db_customer[['country', 'state']]
```

Out[27]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>country</th> <th>state</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | |
|---|---|
| India | Maharashtra |
| India | Karnataka |
| India | Delhi |
| India | Nagaland |
| India | Delhi |
| India | Delhi |
| India | Meghalaya |
| India | Rajasthan |
| Nepal | Kathmandu |
| Nepal | Kathmandu |
| India | Maharashtra |
| India | Karnataka |
| India | Telangana |
| India | Meghalaya |
| India | Uttar Pradesh |
| India | Delhi |
| India | Meghalaya |
| India | Rajasthan |
| India | Telangana |
| India | Uttar Pradesh |
| India | Maharashtra |

In [28]:
```
df_db_customer
```

Out[28]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | | | |
|---|---|---|---|---|---|
| 0002-ORFBO | keshav | India | Maharashtra | Male | 1982-04-12 |
| 0003-MKNFE | raghav | India | Karnataka | Male | 1995-11-23 |
| 0004-TLHLJ | lalita | India | Delhi | Female | 1978-02-15 |
| 0011-IGKFF | mohan | India | Nagaland | Male | 2001-08-30 |
| 0013-EXCHZ | mira | India | Delhi | Female | 1990-05-05 |
| 0013-MHZWF | durga | India | Delhi | Female | 1988-12-10 |
| 0013-SMEOE | mina | India | Meghalaya | Female | 1976-09-21 |
| 0014-BMAQU | madan | India | Rajasthan | Male | 1999-03-14 |
| 0015-UOCOJ | maya | Nepal | Kathmandu | Female | 1985-07-07 |
| 0016-QLJIS | arjun | Nepal | Kathmandu | Male | 1993-10-29 |
| 0017-DINOC | shiva | India | Maharashtra | Male | 1997-01-22 |
| 0017-IUDMW | rangadevi | India | Karnataka | Female | 1981-06-18 |
| 0018-NYROU | chitra | India | Telangana | Female | 2004-12-01 |
| 0019-EFAEP | raju | India | Meghalaya | Female | 1992-04-25 |
| 0019-GFNTW | Madhav | India | Uttar Pradesh | Male | 1979-11-11 |
| 0020-INWCK | parvati | India | Delhi | Female | 1986-02-28 |
| 0020-JDNXP | rikim | India | Meghalaya | Female | 1994-08-19 |
| 0021-IKXGC | vishakha | India | Rajasthan | Female | 2000-09-02 |
| 0022-TCJCI | raghvendra | India | Telangana | Male | 1983-12-30 |
| 0023-HGHWL | rishabh | India | Uttar Pradesh | Male | 1991-05-14 |
| 0023-UYUPN | sudevi | India | Maharashtra | Female | 1977-10-06 |

<h4 id="Second-Table-Cleaning">Second Table Cleaning</h4>

In [29]:
```
df_db_subscription.head()
```

Out[29]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>subscription_start_date</th> <th>subscription_type</th> <th>renewal_date</th> <th>plan_type</th> <th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th> <th>monthly_charges</th> <th>cltv</th> <th>churn_score</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody> </tbody>

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual | NaN | NaN | 13.99 | 627 | 12 |
| 0003-MKNFE | 2020-08-01 | Paid | 2024-08-01 | Premium | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 |
| 0004-TLHLJ | 2022-11-20 | Organic | 2025-11-20 | Basic | Monthly | NaN | NaN | 6.99 | 210 | 34 |
| 0011-IGKFF | 2019-05-10 | Paid | 2025-05-10 | Premium | Annual | NaN | NaN | 22.99 | 1725 | 8 |
| 0013-EXCHZ | 2023-01-05 | Refferal | 2024-01-05 | Standard | Monthly | 2024-02-28 | Too expensive | 13.99 | 195 | 88 |

In [30]:
```
df_db_subscription.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   customerid              21 non-null     str    
 1   subscription_start_date  21 non-null     str    
 2   subscription_type       21 non-null     str    
 3   renewal_date            21 non-null     str    
 4   plan_type               21 non-null     str    
 5   contract_type           21 non-null     str    
 6   cancellation_date       6 non-null      str    
 7   cancellation_reason     6 non-null      str    
 8   monthly_charges         21 non-null     float64
 9   cltv                    21 non-null     int64  
 10  churn_score             21 non-null     int64  
dtypes: float64(1), int64(2), str(8)
memory usage: 1.9 KB
```

In [31]:
```
date_col = ['subscription_start_date', 'renewal_date', 'cancellation_date']
```

In [32]:
```
df_db_subscription[date_col].apply(pd.to_datetime)
```

Out[32]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>subscription_start_date</th> <th>renewal_date</th> <th>cancellation_date</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | |
|---|---|---|---|
| 2021-03-15 | 2025-03-15 | NaT |
| 2020-08-01 | 2024-08-01 | 2024-09-10 |
| 2022-11-20 | 2025-11-20 | NaT |
| 2019-05-10 | 2025-05-10 | NaT |
| 2023-01-05 | 2024-01-05 | 2024-02-28 |
| 2022-06-18 | 2025-06-18 | NaT |
| 2021-09-30 | 2024-09-30 | 2024-11-15 |
| 2020-02-14 | 2025-02-14 | NaT |
| 2023-07-22 | 2024-07-22 | NaT |
| 2022-04-03 | 2025-04-03 | NaT |
| 2021-12-01 | 2025-12-01 | NaT |
| 2023-03-17 | 2024-03-17 | 2024-05-01 |
| 2020-10-09 | 2025-10-09 | NaT |
| 2022-08-25 | 2024-08-25 | 2024-10-31 |
| 2019-11-11 | 2025-11-11 | NaT |
| 2023-05-06 | 2025-05-06 | NaT |
| 2022-01-19 | 2024-01-19 | NaT |
| 2021-07-07 | 2025-07-07 | NaT |
| 2023-09-14 | 2024-09-14 | 2024-09-14 |
| 2020-06-23 | 2025-06-23 | NaT |
| 2022-12-31 | 2025-12-31 | NaT |

In [33]:
```
df_db_subscription[date_col] = df_db_subscription[date_col].apply(pd.to_datetime
df_db_subscription.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customerid              21 non-null     str
 1   subscription_start_date 21 non-null     datetime64[us]
 2   subscription_type       21 non-null     str
 3   renewal_date            21 non-null     datetime64[us]
 4   plan_type               21 non-null     str
 5   contract_type           21 non-null     str
 6   cancellation_date       6 non-null      datetime64[us]
 7   cancellation_reason     6 non-null      str
 8   monthly_charges         21 non-null     float64
 9   cltv                    21 non-null     int64
 10  churn_score             21 non-null     int64
dtypes: datetime64[us](3), float64(1), int64(2), str(5)
memory usage: 1.9 KB
```

<h4 id="Third-Table-Cleaning">Third Table Cleaning</h4>

In [34]:
```
df_db_support.head()
```

Out[34]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> <th>col_1</th> <th>comment</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody> </tbody>

| | | | | | | |
|---|---|---|---|---|---|---|
| 0003-MKNFE | 2024-08-28 00:00:00 | N | 60 | None | service issue |
| 0003-MKNFE | 2024-08-28 00:00:00 | Y | 10 | None | demaned refund |
| 0013-EXCHZ | 2024-01-20 00:00:00 | Y | 20 | None | NaN |
| 0013-MHZWF | 2025-03-18 00:00:00 | N | 90 | None | guidance to renew |
| 0013-SMEOE | 2024-11-01 00:00:00 | N | 30 | None | NaN |

In [35]:
```
df_db_support.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   customerid      9 non-null      str
 1   complaint_date  9 non-null      str
 2   escalations     9 non-null      str
 3   csat_score      9 non-null      int64
 4   col_1           0 non-null      object
 5   comment         4 non-null      str
dtypes: int64(1), object(1), str(4)
memory usage: 564.0+ bytes
```

In [36]:
```
df_db_support.drop(columns = ['col_1', 'comment'], inplace = True)
```

In [37]:
```
df_db_support.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 4 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   customerid      9 non-null      str
 1   complaint_date  9 non-null      str
 2   escalations     9 non-null      str
 3   csat_score      9 non-null      int64
dtypes: int64(1), str(3)
memory usage: 420.0 bytes
```

In [38]:
```
df_db_support['complaint_date'] = pd.to_datetime(df_db_support['complaint_date']

df_db_support.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 4 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   customerid      9 non-null      str
 1   complaint_date  9 non-null      datetime64[us]
 2   escalations     9 non-null      str
 3   csat_score      9 non-null      int64
dtypes: datetime64[us](1), int64(1), str(2)
memory usage: 420.0 bytes
```

<h3 id="3.-Feature-Engineering">3. Feature Engineering</h3>

In [39]:
```
# Create a new column using existing column - churn flag

df_db_subscription['churn_flag'] = np.where(df_db_subscription['cancellation_dat
```

In [40]:
```
df_db_subscription.head()
```

Out[40]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>subscription_start_date</th> <th>subscription_type</th> <th>renewal_date</th> <th>plan_type</th> <th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th> <th>monthly_charges</th> <th>cltv</th> <th>churn_score</th> <th>churn_flag</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual | NaT | NaN | 13.99 | 627 | 12 | 0 |
| 0003-MKNFE | 2020-08-01 | Paid | 2024-08-01 | Premium | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 | 1 |
| 0004-TLHLJ | 2022-11-20 | Organic | 2025-11-20 | Basic | Monthly | NaT | NaN | 6.99 | 210 | 34 | 0 |
| 0011-IGKFF | 2019-05-10 | Paid | 2025-05-10 | Premium | Annual | NaT | NaN | 22.99 | 1725 | 8 | 0 |
| 0013-EXCHZ | 2023-01-05 | Refferal | 2024-01-05 | Standard | Monthly | 2024-02-28 | Too expensive | 13.99 | 195 | 88 | 1 |

In [41]:
```
# Join all the table

(df_db_subscription
    .merge(df_db_customer, on = 'customerid', how = 'left')
    .merge(df_db_support, on = 'customerid', how = 'left'))
```

Out[41]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>subscription_start_date</th> <th>subscription_type</th> <th>renewal_date</th> <th>plan_type</th> <th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th> <th>monthly_charges</th> <th>cltv</th> <th>churn_score</th> <th>churn_flag</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> <th>21</th> <th>22</th> </tbody> <tbody></tbody>

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual | NaT | NaN | 13.99 | 627 | 12 |
| 0003-MKNFE | 2020-08-01 | Paid | 2024-08-01 | Premium | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 |
| 0003-MKNFE | 2020-08-01 | Paid | 2024-08-01 | Premium | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 |
| 0004-TLHLJ | 2022-11-20 | Organic | 2025-11-20 | Basic | Monthly | NaT | NaN | 6.99 | 210 | 34 |
| 0011-IGKFF | 2019-05-10 | Paid | 2025-05-10 | Premium | Annual | NaT | NaN | 22.99 | 1725 | 8 |
| 0013-EXCHZ | 2023-01-05 | Refferal | 2024-01-05 | Standard | Monthly | 2024-02-28 | Too expensive | 13.99 | 195 | 88 |
| 0013-MHZWF | 2022-06-18 | Paid | 2025-06-18 | Standard | Annual | NaT | NaN | 17.99 | 720 | 22 |
| 0013-SMEOE | 2021-09-30 | Refferal | 2024-09-30 | Basic | Monthly | 2024-11-15 | Not enough content | 8.99 | 230 | 79 |
| 0014-BMAQU | 2020-02-14 | Organic | 2025-02-14 | Premium | Annual | NaT | NaN | 22.99 | 1840 | 5 |
| 0015-UOCOJ | 2023-07-22 | Organic | 2024-07-22 | Standard | Monthly | NaT | NaN | 13.99 | 240 | 34 |
| 0016-QLJIS | 2022-04-03 | Organic | 2025-04-03 | Basic | Annual | NaT | NaN | 6.99 | 335 | 41 |
| 0017-DINOC | 2021-12-01 | Organic | 2025-12-01 | Premium | Annual | NaT | NaN | 22.99 | 1610 | 14 |
| 0017-IUDMW | 2023-03-17 | Refferal | 2024-03-17 | Standard | Monthly | 2024-05-01 | Poor streaming quality | 7.99 | 270 | 83 |
| 0018-NYROU | 2020-10-09 | Organic | 2025-10-09 | Standard | Annual | NaT | NaN | 13.99 | 980 | 19 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 0019-EFAEP | 2022-08-25 | Refferal | 2024-08-25 | Basic | Monthly | 2024-10-31 | Switched to competitor | 12.99 | 160 | 76 |
| 0019-GFNTW | 2019-11-11 | Paid | 2025-11-11 | Premium | Annual | NaT | NaN | 92.99 | 2185 | 3 |
| 0020-INWCK | 2023-05-06 | Organic | 2025-05-06 | Standard | Monthly | NaT | NaN | 13.99 | 640 | 58 |
| 0020-JDNXP | 2022-01-19 | Organic | 2024-01-19 | Premium | Annual | NaT | NaN | 20.99 | 550 | 62 |
| 0021-IKXGC | 2021-07-07 | Paid | 2025-07-07 | Standard | Annual | NaT | NaN | 13.99 | 840 | 27 |
| 0022-TCJCI | 2023-09-14 | Refferal | 2024-09-14 | Basic | Monthly | 2024-09-14 | Forgot to cancel trial | 16.99 | 42 | 99 |
| 0022-TCJCI | 2023-09-14 | Refferal | 2024-09-14 | Basic | Monthly | 2024-09-14 | Forgot to cancel trial | 16.99 | 42 | 99 |
| 0023-HGHWL | 2020-06-23 | Organic | 2025-06-23 | Premium | Annual | NaT | NaN | 22.99 | 1955 | 7 |
| 0023-UYUPN | 2022-12-31 | Paid | 2025-12-31 | Standard | Monthly | NaT | NaN | 13.99 | 790 | 47 |

In [42]:
```
df_new = (df_db_subscription
    .merge(df_db_customer, on = 'customerid', how = 'left')
    .merge(df_db_support, on = 'customerid', how = 'left'))
```

In [43]:
```
df_new    # There are duplicates in support table, fix it
```

Out[43]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>subscription_start_date</th> <th>subscription_type</th> <th>renewal_date</th> <th>plan_type</th> <th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th> <th>monthly_charges</th> <th>cltv</th> <th>churn_score</th> <th>churn_flag</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> <th>21</th> <th>22</th> </tbody> <tbody></tbody>

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual | NaT | NaN | 13.99 | 627 | 12 |
| 0003-MKNFE | 2020-08-01 | Paid | 2024-08-01 | Premium | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 |
| 0003-MKNFE | 2020-08-01 | Paid | 2024-08-01 | Premium | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 |
| 0004-TLHLJ | 2022-11-20 | Organic | 2025-11-20 | Basic | Monthly | NaT | NaN | 6.99 | 210 | 34 |
| 0011-IGKFF | 2019-05-10 | Paid | 2025-05-10 | Premium | Annual | NaT | NaN | 22.99 | 1725 | 8 |
| 0013-EXCHZ | 2023-01-05 | Refferal | 2024-01-05 | Standard | Monthly | 2024-02-28 | Too expensive | 13.99 | 195 | 88 |
| 0013-MHZWF | 2022-06-18 | Paid | 2025-06-18 | Standard | Annual | NaT | NaN | 17.99 | 720 | 22 |
| 0013-SMEOE | 2021-09-30 | Refferal | 2024-09-30 | Basic | Monthly | 2024-11-15 | Not enough content | 8.99 | 230 | 79 |
| 0014-BMAQU | 2020-02-14 | Organic | 2025-02-14 | Premium | Annual | NaT | NaN | 22.99 | 1840 | 5 |
| 0015-UOCOJ | 2023-07-22 | Organic | 2024-07-22 | Standard | Monthly | NaT | NaN | 13.99 | 240 | 34 |
| 0016-QLJIS | 2022-04-03 | Organic | 2025-04-03 | Basic | Annual | NaT | NaN | 6.99 | 335 | 41 |
| 0017-DINOC | 2021-12-01 | Organic | 2025-12-01 | Premium | Annual | NaT | NaN | 22.99 | 1610 | 14 |
| 0017-IUDMW | 2023-03-17 | Refferal | 2024-03-17 | Standard | Monthly | 2024-05-01 | Poor streaming quality | 7.99 | 270 | 83 |
| 0018-NYROU | 2020-10-09 | Organic | 2025-10-09 | Standard | Annual | NaT | NaN | 13.99 | 980 | 19 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 0019-EFAEP | 2022-08-25 | Refferal | 2024-08-25 | Basic | Monthly | 2024-10-31 | Switched to competitor | 12.99 | 160 | 76 |
| 0019-GFNTW | 2019-11-11 | Paid | 2025-11-11 | Premium | Annual | NaT | NaN | 92.99 | 2185 | 3 |
| 0020-INWCK | 2023-05-06 | Organic | 2025-05-06 | Standard | Monthly | NaT | NaN | 13.99 | 640 | 58 |
| 0020-JDNXP | 2022-01-19 | Organic | 2024-01-19 | Premium | Annual | NaT | NaN | 20.99 | 550 | 62 |
| 0021-IKXGC | 2021-07-07 | Paid | 2025-07-07 | Standard | Annual | NaT | NaN | 13.99 | 840 | 27 |
| 0022-TCJCI | 2023-09-14 | Refferal | 2024-09-14 | Basic | Monthly | 2024-09-14 | Forgot to cancel trial | 16.99 | 42 | 99 |
| 0022-TCJCI | 2023-09-14 | Refferal | 2024-09-14 | Basic | Monthly | 2024-09-14 | Forgot to cancel trial | 16.99 | 42 | 99 |
| 0023-HGHWL | 2020-06-23 | Organic | 2025-06-23 | Premium | Annual | NaT | NaN | 22.99 | 1955 | 7 |
| 0023-UYUPN | 2022-12-31 | Paid | 2025-12-31 | Standard | Monthly | NaT | NaN | 13.99 | 790 | 47 |

In [44]:
```
df_new.shape   # Here 2 duplicate rows has been added
```

Out[44]:
```
(23, 20)
```

In [45]:
```
df_db_subscription.shape   # there are 21 records in subs table
```

Out[45]:
```
(21, 12)
```

In [46]:
```
df_db_subscription['customerid'].nunique()     # it shows total unique value
```

Out[46]:
```
21
```

In [47]:
```
df_db_customer['customerid'].nunique()
```

Out[47]:
```
21
```

In [48]:
```
df_db_support['customerid'].nunique()
```

Out[48]:
```
7
```

In [49]:
```
df_db_support['customerid'].size
```

Out[49]:
```
9
```

In [50]:
```
df_db_support
```

Out[50]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> </tbody> <tbody></tbody>

| | | | | |
|---|---|---|---|---|
| 0003-MKNFE | 2024-08-28 | N | 60 | |
| 0003-MKNFE | 2024-08-28 | Y | 10 | |
| 0013-EXCHZ | 2024-01-20 | Y | 20 | |
| 0013-MHZWF | 2025-03-18 | N | 90 | |
| 0013-SMEOE | 2024-11-01 | N | 30 | |
| 0017-IUDMW | 2024-04-10 | Y | 25 | |
| 0019-EFAEP | 2024-09-27 | Y | 30 | |
| 0022-TCJCI | 2024-09-13 | Y | 10 | |
| 0022-TCJCI | 2024-09-14 | N | 90 | |

In [51]:
```
df_db_support['complaint_count'] = df_db_support.groupby('customerid')['customer
```

In [52]:
```
df_db_support
```

Out[52]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> <th>complaint_count</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> </tbody> <tbody></tbody>

| | | | | | |
|---|---|---|---|---|---|
| 0003-MKNFE | 2024-08-28 | N | 60 | | 2 |
| 0003-MKNFE | 2024-08-28 | Y | 10 | | 2 |
| 0013-EXCHZ | 2024-01-20 | Y | 20 | | 1 |
| 0013-MHZWF | 2025-03-18 | N | 90 | | 1 |
| 0013-SMEOE | 2024-11-01 | N | 30 | | 1 |
| 0017-IUDMW | 2024-04-10 | Y | 25 | | 1 |
| 0019-EFAEP | 2024-09-27 | Y | 30 | | 1 |
| 0022-TCJCI | 2024-09-13 | Y | 10 | | 2 |
| 0022-TCJCI | 2024-09-14 | N | 90 | | 2 |

In [53]:
```
df_db_support.sort_values('complaint_date').drop_duplicates('customerid', keep =
```

Out[53]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> <th>complaint_count</th> </thead> <tbody> <th>2</th> <th>5</th> <th>1</th> <th>8</th> <th>6</th> <th>4</th> <th>3</th> </tbody> <tbody></tbody>

| | | | | | |
|---|---|---|---|---|---|
| 0013-EXCHZ | 2024-01-20 | Y | 20 | 1 | |
| 0017-IUDMW | 2024-04-10 | Y | 25 | 1 | |
| 0003-MKNFE | 2024-08-28 | Y | 10 | 2 | |
| 0022-TCJCI | 2024-09-14 | N | 90 | 2 | |
| 0019-EFAEP | 2024-09-27 | Y | 30 | 1 | |
| 0013-SMEOE | 2024-11-01 | N | 30 | 1 | |
| 0013-MHZWF | 2025-03-18 | N | 90 | 1 | |

In [54]:
```
df_db_support = df_db_support.sort_values('complaint_date').drop_duplicates('cus
```

In [55]:
```
df_db_support
```

Out[55]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> <th>complaint_count</th> </thead> <tbody> <th>2</th> <th>5</th> <th>1</th> <th>8</th> <th>6</th> <th>4</th> <th>3</th> </tbody> <tbody></tbody>

| | | | | | |
|---|---|---|---|---|---|
| 0013-EXCHZ | 2024-01-20 | Y | 20 | 1 | |
| 0017-IUDMW | 2024-04-10 | Y | 25 | 1 | |
| 0003-MKNFE | 2024-08-28 | Y | 10 | 2 | |
| 0022-TCJCI | 2024-09-14 | N | 90 | 2 | |
| 0019-EFAEP | 2024-09-27 | Y | 30 | 1 | |
| 0013-SMEOE | 2024-11-01 | N | 30 | 1 | |
| 0013-MHZWF | 2025-03-18 | N | 90 | 1 | |

In [56]:
```
df_db_support['customerid'].size
```

Out[56]: 7

In [57]:
```
# merge

df = (df_db_subscription
    .merge(df_db_customer, on = 'customerid', how = 'left')
    .merge(df_db_support, on = 'customerid', how = 'left'))
```

In [58]:
```
df.shape
```

Out[58]: (21, 21)

<h3 id="4.-Data-Analysis">4. Data Analysis</h3>

In [59]:
```
# 1. Churn Rate
```

In [60]:
```
df.columns
```

Out[60]:
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count'],
      dtype='str')
```

In [61]:
```
churn_rate = df['churn_flag'].mean()*100
print("churn_rate is = ", round(churn_rate, 2), "%")
```

```
churn_rate is =  28.57 %
```

In [62]:
```
# 2. Retention Rate

Retention_rate = 100 - churn_rate
print("Retention Rate is = ", round(Retention_rate, 2), "%")
```

```
Retention Rate is =  71.43 %
```

In [63]:
```
# 3. Churn by plan type

churn_by_plan = (df.groupby('plan_type')['churn_flag'].mean().mul(100).round(2).
print(churn_by_plan)
```

```
  plan_type  churn_rate_percent
0     Basic               60.00
1   Premium               14.29
2  Standard               22.22
```

In [64]:
```
# 5. Avg Revenue Per User

arpu = df['monthly_charges'].mean()
print("ARPU = ", round(arpu, 2))
```

```
ARPU =  18.85
```

In [65]:
```
# 6. Avg Customer Tenure

# count of days users have used our service : cancellation date else current dat

today = pd.Timestamp.today()

# formula = np.where(condition, if True, if False)

df['tenure_days'] = np.where(
                            df['cancellation_date'].notna(),
                            (df['cancellation_date'] - df['subscription_start_da
                            (today - df['subscription_start_date']).dt.days
                            )

avg_tenure = df['tenure_days'].mean()
print("Avg Tenure in days = ", round(avg_tenure))
```

```
Avg Tenure in days =  1552
```

In [66]:
```
# 7. Revenue at risk = revenue lost from churned users

revenue_at_risk = df.loc[df['churn_flag']==1, 'monthly_charges'].sum()
print("Revenue at risk (Rs 'K') = ", revenue_at_risk)
```

```
Revenue at risk (Rs 'K') =  73.94
```

In [67]:
```
# 8. Escalation rate

escalation_rate = (df['escalations']=='Y').mean()*100
print("Escalation Rate = ", round(escalation_rate, 2), "%")
```

```
Escalation Rate =  19.05 %
```

In [68]:
```
# 9. Avg complaint per user

avg_complaints = df['complaint_count'].sum() / df['customerid'].nunique()
print("Avg complaint per user = ", round(avg_complaints,2))
```

```
Avg complaint per user =  0.43
```

In [69]:
```
# 10. Correlation Escalation vs Churn

df['escalations'] = np.where(df['escalations'] == 'Y',1, 0)    # encoding string

corr_df = df[['escalations', 'churn_flag']].dropna()

correlation = corr_df['escalations'].corr(df['churn_flag'])
print("Correlation between escalation and churn is = ", round(correlation, 2))
```

```
Correlation between escalation and churn is =  0.77
```

In [70]:
```
# 11. Create a column using existing column - Churn Risk

conditions = [
                (df['churn_score'] < 50),
                (df['churn_score'] >= 50) & (df['churn_score'] < 70),
                (df['churn_score'] >= 70)
            ]

choices = ['Low', 'Med', 'High']

df['Churn_Risk'] = np.select(conditions, choices, default = 'Unknown')
```

In [71]:
```
df[['churn_score', 'Churn_Risk']]
```

Out[71]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>churn_score</th> <th>Churn_Risk</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | |
|---|---|
| 12 | Low |
| 91 | High |
| 34 | Low |
| 8 | Low |
| 88 | High |
| 22 | Low |
| 79 | High |
| 5 | Low |
| 34 | Low |
| 41 | Low |
| 14 | Low |
| 83 | High |
| 19 | Low |
| 76 | High |
| 3 | Low |
| 58 | Med |
| 62 | Med |
| 27 | Low |
| 99 | High |
| 7 | Low |
| 47 | Low |

<h3 id="5.-Visualization-using-Matplotlib">5. Visualization using Matplotlib</h3>

In [72]:
```
df_visual = df.copy()
```

In [73]:
```
df_visual.head()
```

Out[73]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>customerid</th> <th>subscription_start_date</th> <th>subscription_type</th> <th>renewal_date</th> <th>plan_type</th> <th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th> <th>monthly_charges</th> <th>cltv</th> <th>...</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> <th>complaint_count</th> <th>tenure_days</th> <th>Churn_Risk</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

| | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual | NaT | NaN | 13.99 | 627 | ... | In |
| 0003-MKNFE | 2020-08-01 | Paid | 2024-08-01 | Premium | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | ... | In |
| 0004-TLHLJ | 2022-11-20 | Organic | 2025-11-20 | Basic | Monthly | NaT | NaN | 6.99 | 210 | ... | In |
| 0011-IGKFF | 2019-05-10 | Paid | 2025-05-10 | Premium | Annual | NaT | NaN | 22.99 | 1725 | ... | In |
| 0013-EXCHZ | 2023-01-05 | Refferal | 2024-01-05 | Standard | Monthly | 2024-02-28 | Too expensive | 13.99 | 195 | ... | In |

5 rows × 23 columns

In [74]:
```
df_visual.shape
```

Out[74]: (21, 23)

In [75]:
```
df_visual.columns
```

Out[75]:
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'tenure_days', 'Churn_Risk'],
      dtype='str')
```

In [76]:
```
# 5.1 Monthly Churn Trend ( Time series KPI )

df_visual['cancellation_month'] = df_visual['cancellation_date'].dt.to_period('M

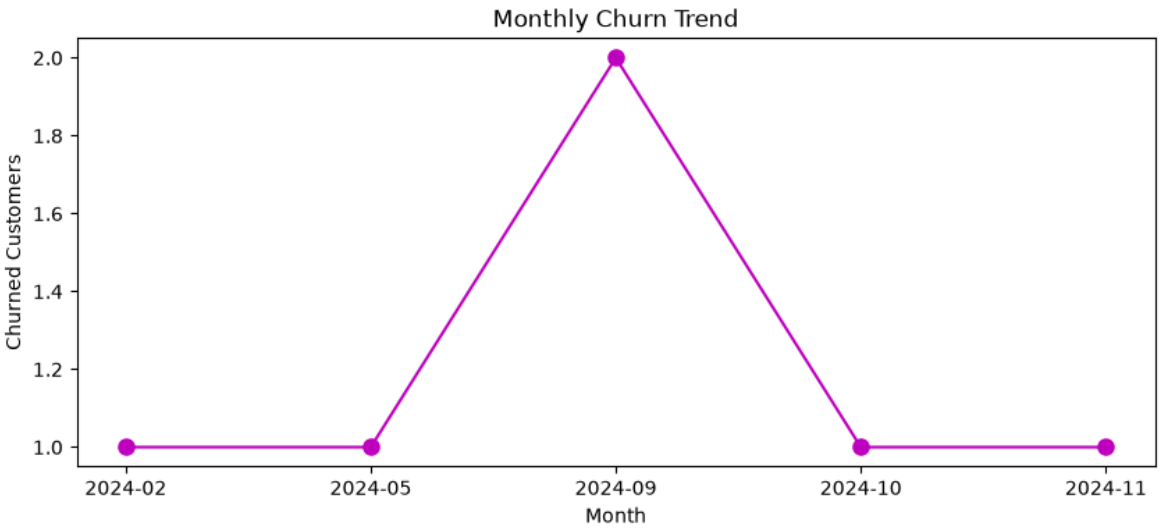
churn_trend = df_visual[df_visual['churn_flag'] == 1].groupby('cancellation_mont

plt.figure(figsize = (10,4))

plt.plot(churn_trend.index.astype(str), churn_trend.values, color='m', marker='o

plt.title("Monthly Churn Trend")
plt.xlabel('Month')
```

```
plt.ylabel('Churned Customers')
plt.show()
```




In [77]:
```
# 5.2 Churn by plan type

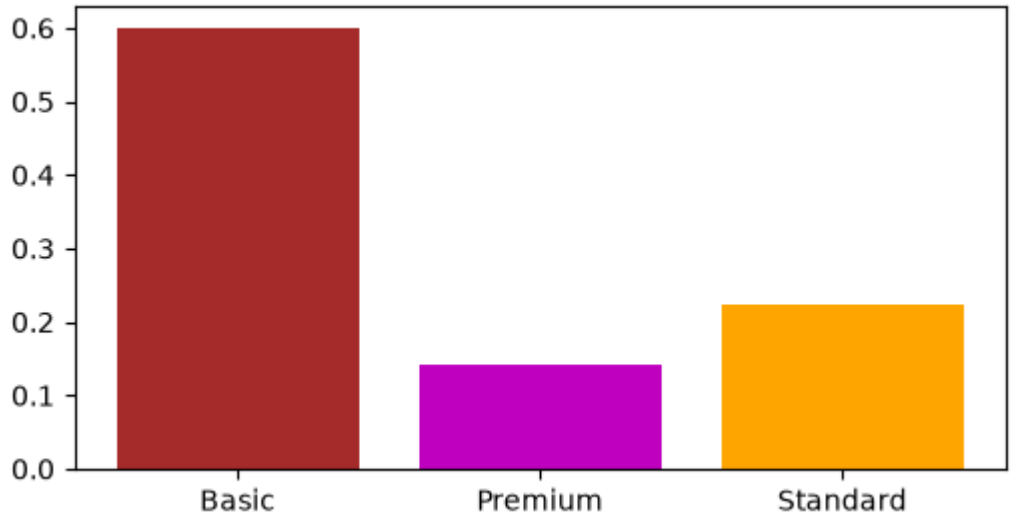
churn_plan = df_visual.groupby('plan_type')['churn_flag'].mean()

# we can give manual color to each bar.
colors = ['brown', 'm', 'orange']

# automatic color assignment
# colors = plt.cm.Set2(np.linspace(0, 1, len(churn_plan)))

plt.figure(figsize = (6,3))

plt.bar(churn_plan.index, churn_plan.values, color = colors)
plt.show()
```




In [78]:
```
# 5.3 Churn by State

churn_plan = df_visual.groupby('state')['churn_flag'].mean()

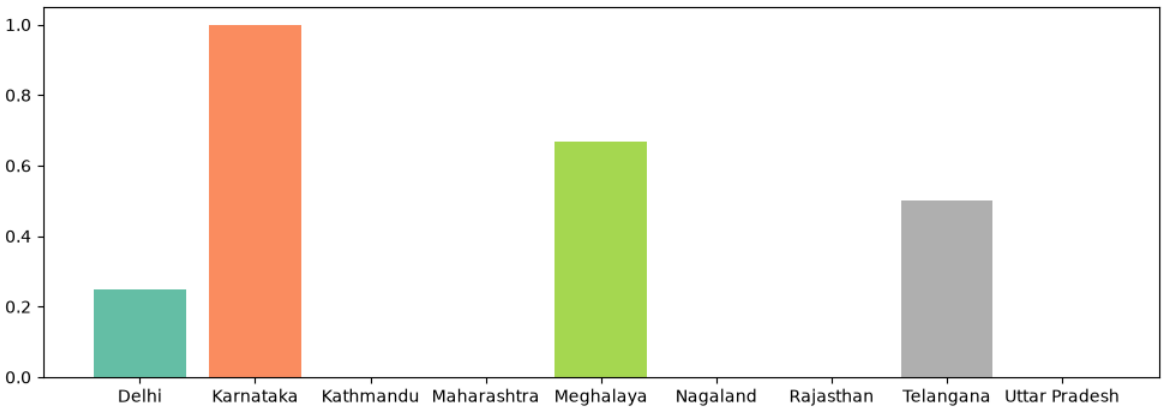
# we can give manual color to each bar.
# colors = ['brown', 'm', 'orange']

# automatic color assignment
```

```
colors = plt.cm.Set2(np.linspace(0, 1, len(churn_plan)))

plt.figure(figsize = (12,4))

plt.bar(churn_plan.index, churn_plan.values, color = colors)
plt.show()
```




<h4 id="Visualization-Using-Seaborn">Visualization Using Seaborn</h4>

In [79]:
```
# encoding - covert string to integer so that we can find correlation between fe
df_visual.columns
```

Out[79]:
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'tenure_days', 'Churn_Risk', 'cancellation_month'],
      dtype='str')
```

In [80]:
```
df_visual[['plan_type', 'contract_type', 'churn_score', 'churn_flag', 'Churn_Ris
```

Out[80]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>plan_type</th> <th>contract_type</th> <th>churn_score</th> <th>churn_flag</th> <th>Churn_Risk</th> <th>escalations</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody> </tbody>

| | | | | | | |
|---|---|---|---|---|---|---|
| Standard | Annual | 12 | 0 | Low | 0 | |
| Premium | Annual | 91 | 1 | High | 1 | |
| Basic | Monthly | 34 | 0 | Low | 0 | |
| Premium | Annual | 8 | 0 | Low | 0 | |
| Standard | Monthly | 88 | 1 | High | 1 | |

In [81]:
```
# Incorrect method of encoding - as numbers are not assigned based on priority

df_encoded = df_visual[['plan_type', 'contract_type', 'churn_score', 'churn_flag

categorial_cols = ['plan_type', 'contract_type', 'Churn_Risk']
```

```
for col in categorial_cols:
    df_encoded[col] = df_encoded[col].astype('category').cat.codes
```

In [82]:
```
df_encoded.head()
```

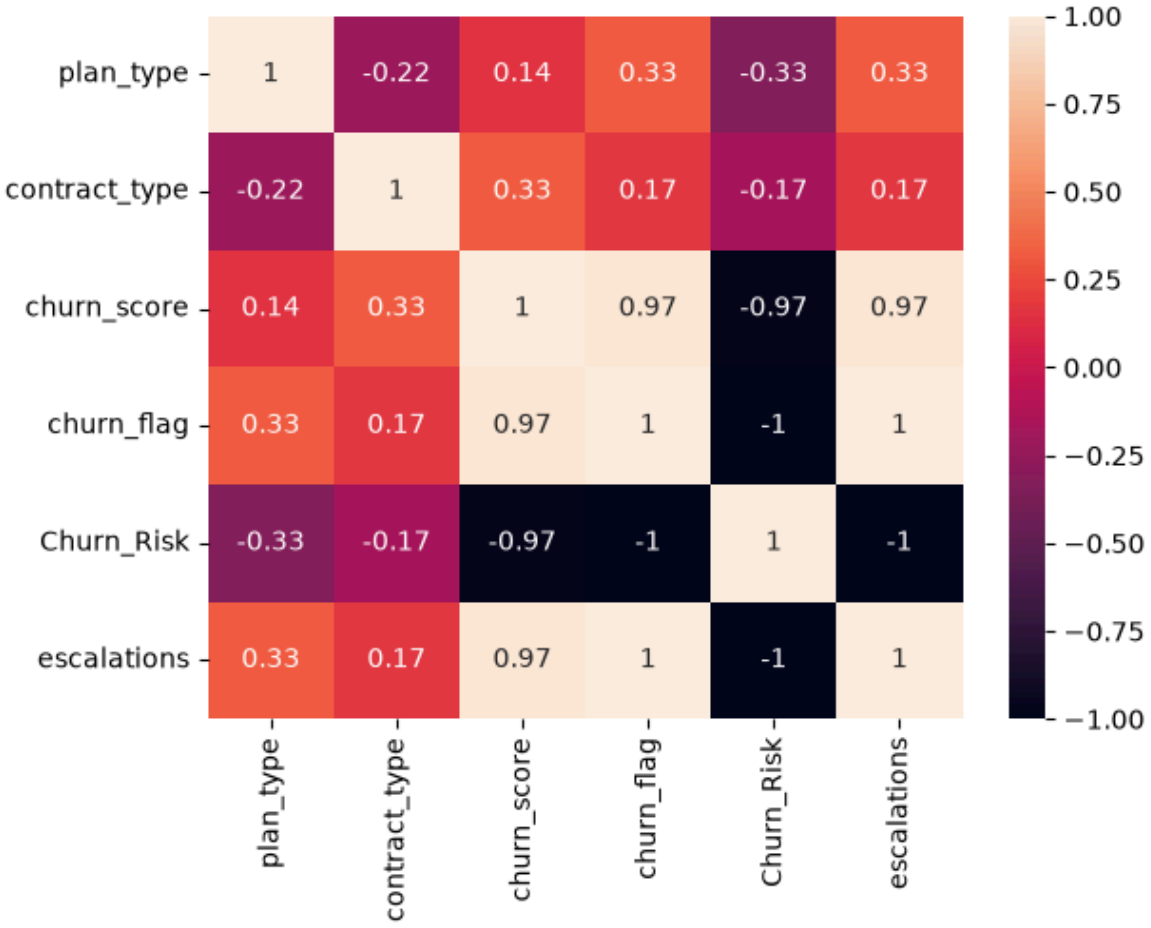
Out[82]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>plan_type</th> <th>contract_type</th> <th>churn_score</th> <th>churn_flag</th> <th>Churn_Risk</th> <th>escalations</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody> </tbody>

| | | | | | | |
|---|---|---|---|---|---|---|
| 2 | 0 | 12 | 0 | 1 | 0 | |
| 1 | 0 | 91 | 1 | 0 | 1 | |
| 0 | 1 | 34 | 0 | 1 | 0 | |
| 1 | 0 | 8 | 0 | 1 | 0 | |
| 2 | 1 | 88 | 1 | 0 | 1 | |

In [83]:
```
# Heatmap (correlation matrix)

sns.heatmap(df_encoded.corr(), annot=True)
```

Out[83]: <Axes: >




In [84]:
```
# Correct method of encoding - based on priority
```

```
df_encoded = df_visual[['plan_type', 'contract_type', 'churn_score', 'churn_flag

order_mappings = {
    'plan_type': ['Basic', 'Standard', 'Premium'],
    'contract_type' :  ['Monthly',  'Annual'],
    'Churn_Risk' : ['Low', 'Med', 'High']
}

for col, order in order_mappings.items():
    df_encoded[col] = pd.Categorical(df_encoded[col].astype('category'), categor
```

In [85]:
```
df_visual[['plan_type', 'contract_type', 'churn_score', 'churn_flag', 'Churn_Ris
```

Out[85]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>plan_type</th> <th>contract_type</th> <th>churn_score</th> <th>churn_flag</th> <th>Churn_Risk</th> <th>escalations</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody> </tbody>

| | | | | | |
|---|---|---|---|---|---|
| Standard | Annual | 12 | 0 | Low | 0 |
| Premium | Annual | 91 | 1 | High | 1 |
| Basic | Monthly | 34 | 0 | Low | 0 |
| Premium | Annual | 8 | 0 | Low | 0 |
| Standard | Monthly | 88 | 1 | High | 1 |

In [86]:
```
df_encoded
```

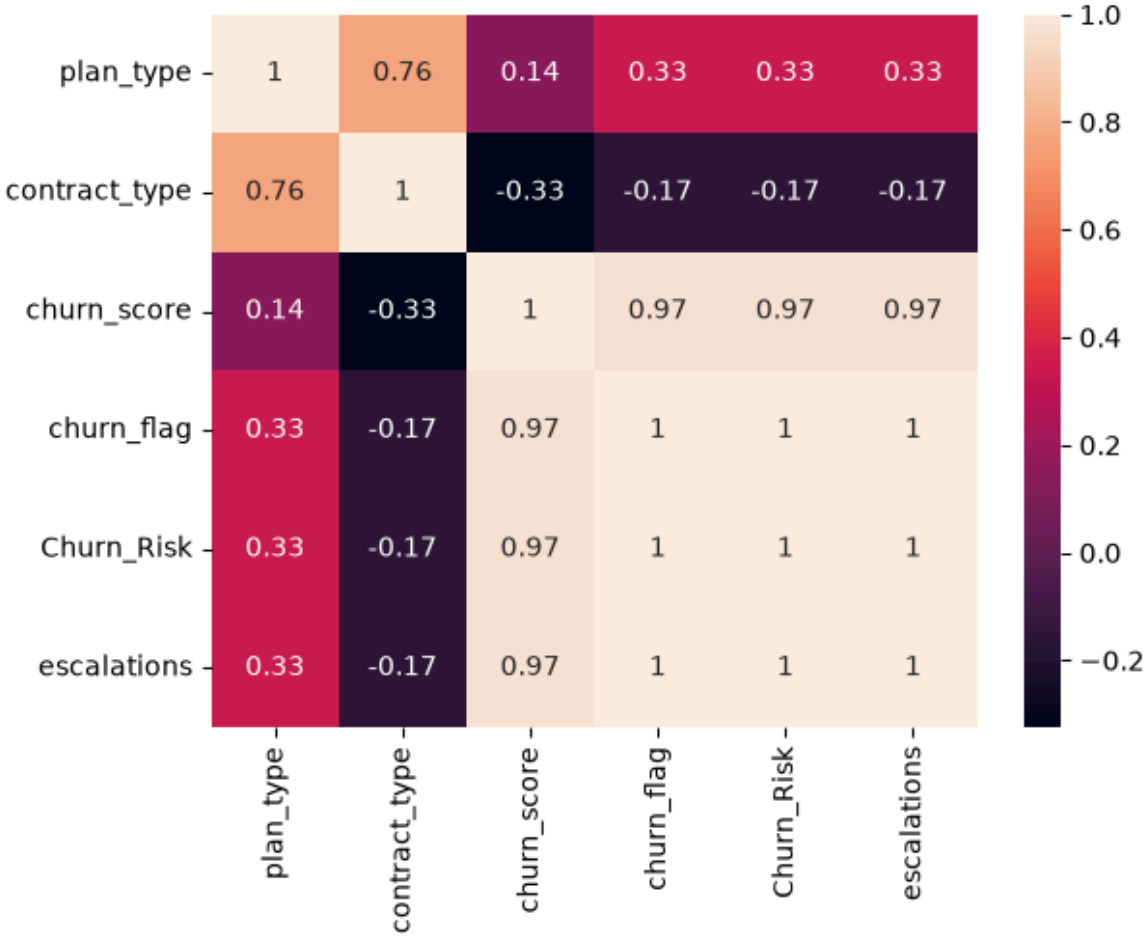
Out[86]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>plan_type</th> <th>contract_type</th> <th>churn_score</th> <th>churn_flag</th> <th>Churn_Risk</th> <th>escalations</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody> </tbody>

| | | | | | |
|---|---|---|---|---|---|
| 1 | 1 | 12 | 0 | 0 | 0 |
| 2 | 1 | 91 | 1 | 2 | 1 |
| 0 | 0 | 34 | 0 | 0 | 0 |
| 2 | 1 | 8 | 0 | 0 | 0 |
| 1 | 0 | 88 | 1 | 2 | 1 |

In [87]:
```
# Heatmap (correlation matrix)

sns.heatmap(df_encoded.corr(), annot=True)
```

Out[87]: <Axes: >

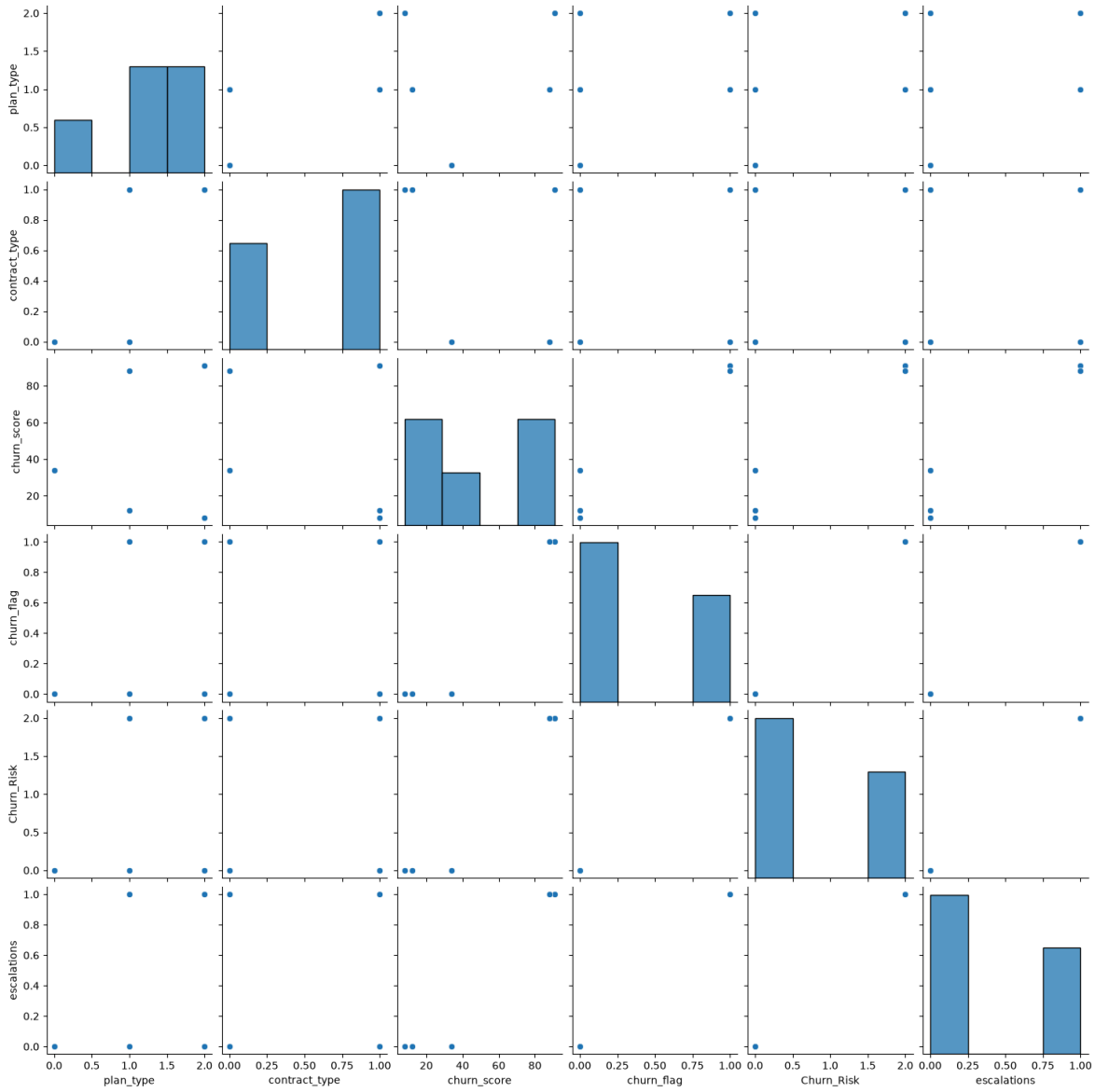



<h4 id="Seaborn-Pairplot">Seaborn Pairplot</h4>

In [88]:
```
# pairplot = relationship in a dataset

sns.pairplot(df_encoded)
```

Out[88]: <seaborn.axisgrid.PairGrid at 0x1feaa3fb380>

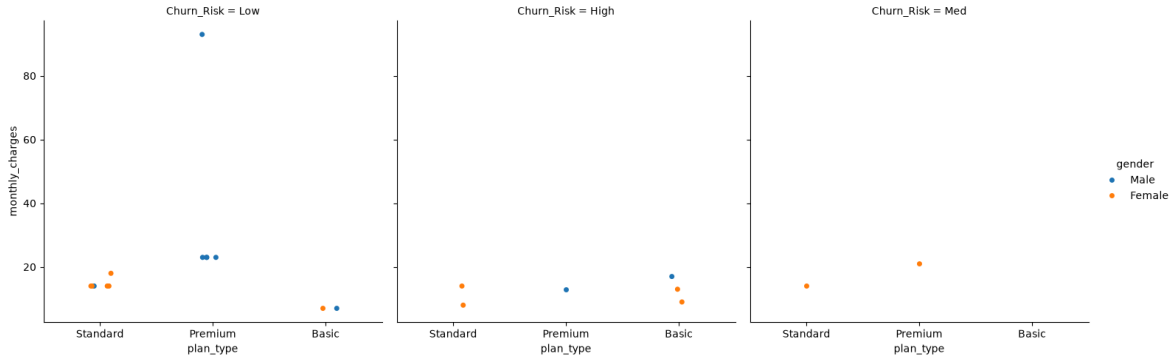



<h4 id="Categorial-Plot">Categorial Plot</h4>

In [89]:
```
# catplot/facegrid plot - multi-dim comparison

sns.catplot(data=df_visual,
            x = 'plan_type',
            y = 'monthly_charges',
            hue = 'gender',
            col = 'Churn_Risk' )
```

Out[89]: <seaborn.axisgrid.FacetGrid at 0x1fead19ee40>




<h3 id="Pivot-Table">Pivot Table</h3>

In [90]:
```
pd.pivot_table(df_visual,
               index = 'plan_type',
               values = 'churn_flag',
               aggfunc = 'mean' )
```

Out[90]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>churn_flag</th> <th>plan_type</th> <th></th> </thead> <tbody> <th>Basic</th> <th>Premium</th> <th>Standard</th> </tbody> <tbody></tbody>

0.600000

0.142857

0.222222

In [91]:
```
df_visual.columns
```

Out[91]:
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'tenure_days', 'Churn_Risk', 'cancellation_month'],
      dtype='str')
```

In [92]:
```
pd.pivot_table(df_visual,
               index = 'plan_type',
               values = ['monthly_charges', 'customerid', 'churn_flag'],
               aggfunc = {'monthly_charges' : 'sum',
                          'customerid' : 'nunique',
                          'churn_flag' : 'mean'
                         }
              )
```

Out[92]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>churn_flag</th> <th>customerid</th> <th>monthly_charges</th> <th>plan_type</th> <th></th> <th></th> <th></th> </thead> <tbody> <th>Basic</th> <th>Premium</th> <th>Standard</th> </tbody> <tbody></tbody>

0.600000 5 52.95

0.142857 7 218.93

0.222222 9 123.91

<h3 id="SQL-in-Python-(Pandas)">SQL in Python (Pandas)</h3>

In [93]:
```
# Create db in sql

conn = sqlite3.connect('test_database.sqlite')

# table details
conn.execute("create table users (first_name TEXT, country TEXT, budget INTEGER)
```

```
# commit and save
conn.commit()
```

In [94]:
```
# inset data

cursor = conn.cursor()

cursor.execute(
    """
        insert into users values ('Saksham', 'Germany', 6000),
                                 ('Payal', 'Italy', 7000),
                                 ('Santosh', 'India', 12000),
                                 ('Sonam', 'India', 9000)
    """
)

conn.commit()
```

In [98]:
```
# check inserted data in table

conn = sqlite3.connect('test_database.sqlite')

#df_result = pd.read_sql(""" SELECT * FROM users """, conn)

# OR

query = """ SELECT * FROM users """

df_result = pd.read_sql(query, conn)

df_result
```

Out[98]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>first_name</th> <th>country</th> <th>budget</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> </tbody> <tbody> </tbody>

| | | | |
|---|---|---|---|
| Saksham | Germany | 6000 | |
| Payal | Italy | 7000 | |
| Santosh | India | 12000 | |
| Sonam | India | 9000 | |

In [99]:
```
# Aggregation

query = """ SELECT COUNTRY, sum(budget) from users group by country """

df_aggr = pd.read_sql(query, conn)

df_aggr
```

Out[99]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>country</th> <th>sum(budget)</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> </tbody> <tbody></tbody>

| | |
|---|---|
| Germany | 6000 |
| India | 21000 |
| Italy | 7000 |

In [100…

```
# Always close the connection of database once the task is over

conn.close()
```

In [ ]: